Stat 517
Homework # 1

1. For each of the following, find the constant c so that $p(x)$ satisfies the condition of being a probability mass function (pmf) of one random variable X.

   (a) $p(x) = c(\frac{2}{3})^x, x = 1, 2, 3, \ldots$, zero elsewhere.

   (b) $p(x) = cx$, $x = 1, 2, 3, 4, 5, 6$, zero elsewhere.

2. If the probability density function (pdf) of X is $f(x) = 2xe^{-x^2}$, $0 < x < \infty$, zero elsewhere, determine the pdf of $Y = X^2$.

3. Let X have the pdf $f(x) = 3x^2$, $0 < x < 1$, zero elsewhere.

   (a) Compute $\mathbb{E}(X^3)$.

   (b) Show that $Y = X^3$ has a uniform$(0, 1)$ distribution.

   (c) Compute $E(Y)$ and compare this result with the answer obtained in part (a).

4. Let $p(x) = (\frac{1}{2})^x$, $x = 1, 2, 3, \ldots$, zero elsewhere, be the pmf of the random variable X. Find the moment generating function (mgf), the mean, and the variance of X.

5. Let $f(x_1, x_2) = 4x_1x_2$, $0 < x_1 < 1$, $0 < x_2 < 1$, zero elsewhere, be the pdf of X_1 and X_2. Find $P(0 < X_1 < \frac{1}{2}, \frac{1}{4} < X_2 < 1)$, $P(X_1 = X_2)$, $P(X_1 < X_2)$, and $P(X_1 \leq X_2)$.

6. Let $f(x, y) = e^{-x-y}$, $0 < x < \infty$, $0 < y < \infty$, zero elsewhere, be the pdf of X and Y. Then if $Z = X + Y$, compute $P(X \leq 0)$, $P(Z \leq 6)$, and more generally $P(Z \leq z)$ for $0 < z < \infty$. What is the pdf of Z?

7. Let X_1 and X_2 have the joint pdf $f(x_1, x_2) = 15x_1^2x_2$, $0 < x_1 < x_2 < 1$, zero elsewhere. Find the marginal pdfs and compute $P(X_1 + X_2 \leq 1)$.

8. Let X and Y have the joint pdf $f(x, y) = 3x$, $0 < y < x < 1$, zero elsewhere. Are X and Y independent? If not, find $E(X|y)$.